

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Regular & Supplementary Winter Examination-2023

Course: B. Tech. CSE(AI&ML) / AI&DS

Semester : VII

Subject Code & Name: BTAIC701 Natural Language Processing

Max Marks: 60

Date:02-01-2024

Duration: 3 Hr.

Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

(Level/CO) Marks

Q. 1	Solve Any Two of the following.		12
A)	What is Natural Language Processing? Discuss with some applications.	Understand	6
B)	What is the purpose of text normalization in natural language processing and Explain the three common tasks in text normalization?	Understand	6
C)	Explain the "Minimum Edit Distance Algorithm" and compute the minimum edit distance between EXECUTION and INTENTION	Apply	6
Q.2	Solve Any Two of the following.		12
A)	What is a language model? Write note on N-Gram language model.	Understand	6
B)	Find the probability of test sentence given below if a bigram model is used on the following training sentences	Apply	6
	Training data:		
	1.The Arabian Knights		
	2.These are the fairy tales of the eat		
	3.The Stories of the Arabian Knights are translated primary languages.		
	Test Data: The Arabian Knights are the fairy tales of the eat		
C)	Explain vector space model of information retrieval		6
Q. 3	Solve Any Two of the following.		12
A)	Perform parsing using simple top down parsing for the sentence "The dogs cried" using the grammar given below:	Apply	6
	S->NP VP		
	NP->ART N		
	NP->ART ADJ N		
	VP->V		

VP->V NP

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|--|-------------------|----------|
| B) Show working of NER with appropriate example. | Understand | 6 |
| C) Illustrate maximum entropy principle formula with example. | Understand | 6 |

Q.4 Solve Any Two of the following. **12**

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|---|-------------------|----------|
| A) Write a short note on importance of sentiment analysis. | Understand | 6 |
| B) Apply the Logistic Regression model to perform text classification. | Apply | 6 |
| C) What is Text Classification? Explain How Does Text Classification Work? | Understand | 6 |

Q. 5 Solve Any Two of the following. **12**

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|---|-------------------|----------|
| A) Explain text summarization and multiple document text summarization with neat diagram | Understand | 6 |
| B) Explain the simple RNN with its application | Understand | 6 |
| C) What is textual entailment in NLP? Apply it on question answering application and simplify its working with suitable example. | Apply | 6 |

***** End *****